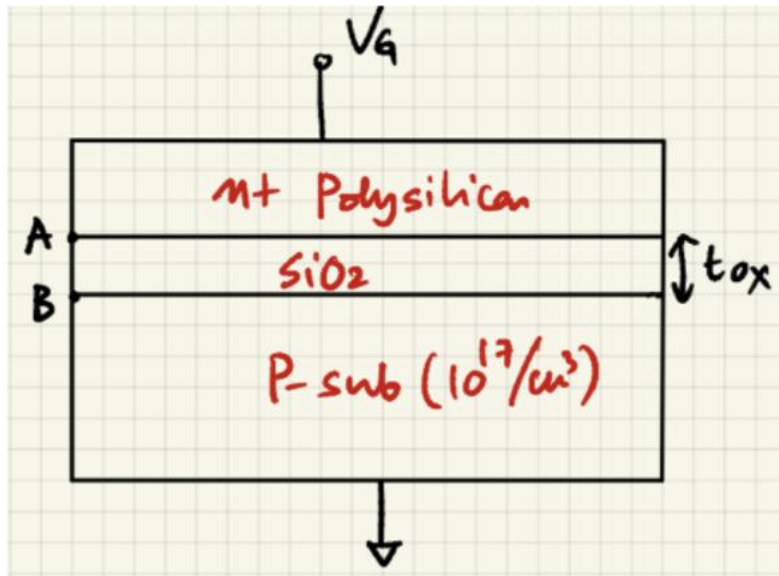


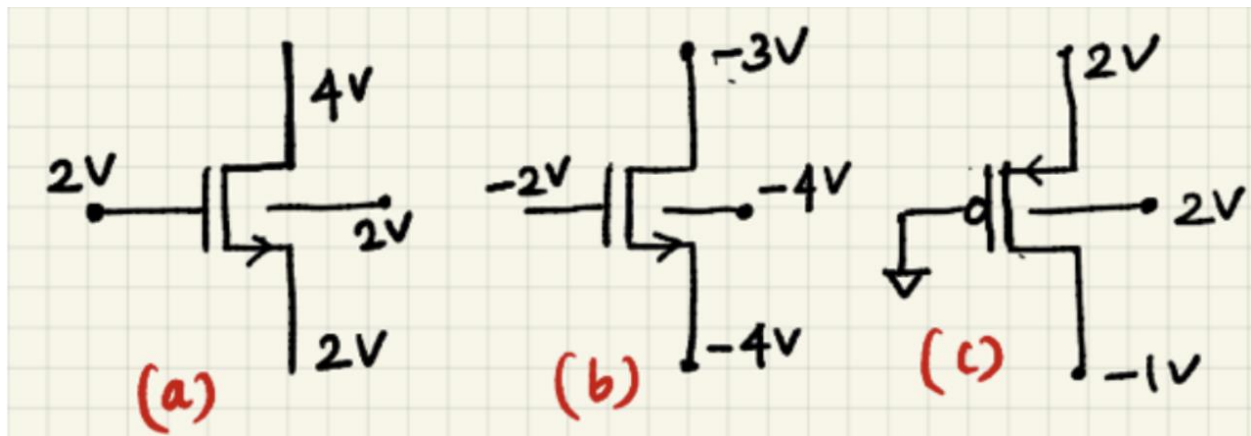
Indian Institute of Technology, Gandhinagar
 ES226 Semiconductor Devices
 Semester 2
 Tutorial – 9

Q1: The Figure shows a MOS capacitor. At a particular V_G , there are depletion regions at Poly-Silicon/ SiO_2 and SiO_2/Si interfaces (A and B). If the depletion width in poly-silicon (W_P) is 9nm and depletion width in silicon (W_D) is 30nm.

- Find the total capacitance of the device.
- What is the electric field in oxide.



Q2. Find the region of operation of transistors shown in Figure ($V_{thn}=|V_{thp}|=1\text{V}$)



Q3. Draw an transfer characteristics of NMOS transistor for a different substrate doping concentration.

Q4. Sketch the following curves and explain.

- a) V_{th} vs L_G
- b) V_{th} vs t_{ox}
- c) Subthreshold Swing vs L_G
- d) Subthreshold Swing vs t_{ox}
- e) Subthreshold Swing vs temperature

Q5. An n-type MOS capacitor shows a “stretch-out” in its C–V curve but no hysteresis.

- a) Explain the physical reason behind this behavior.
- b) Derive the relation between interface trap capacitance C_{it} and interface trap density D_{it} .
- c) How does an increase in D_{it} affect the subthreshold swing of a MOSFET?

Q6. Derive the subthreshold slope (SS) for an n-MOSFET starting from the Boltzmann relation for channel charge. Then compute SS (mV/decade) at 300 K if $C_{ox}' = 3.45 \times 10^{-7}$ F/cm² and the depletion capacitance per unit area near weak inversion is $C_{dep}' = 5.0 \times 10^{-7}$ F/cm².

Q7. An n-MOSFET is fabricated on a p-type silicon substrate with a doping concentration of $N_A = 2 \times 10^{16}$ cm⁻³. The oxide thickness is $t_{ox} = 15$ nm, and the threshold voltage at zero substrate bias ($V_{SB} = 0$) is $V_{T0} = 0.5$ V. Assume $T = 300$ K and $n_i = 1.5 \times 10^{10}$ cm⁻³.

- a) Derive the expression for the threshold voltage V_T as a function of substrate bias V_{SB} .
- b) Calculate the threshold voltage when $V_{SB} = 2$ V.
- c) Calculate the maximum depletion width at $V_{SB} = 0$ and at $V_{SB} = 2$ V.

Q8. A MOSFET has an oxide capacitance of $C_{ox} = 2 \times 10^{-7}$ F/cm² and a depletion capacitance of $C_{dep} = 1 \times 10^{-7}$ F/cm².

- a) Explain why the subthreshold slope of a real MOSFET is greater than the ideal value.
- b) Calculate the subthreshold slope (SS) at $T = 300$ K.

Also, state the ideal subthreshold slope at 300 K.

Q9. An NFET is made with $t_{ox} = 4$ nm, $L = 1$ μ m, $W = 2$ μ m, $V_T = 1$ V, and $\mu_f = 500$ cm² / V · s. If the simple model is used, what should the width of the PFET be to get the same characteristics (apart from polarity). Let the low field mobility for holes be 200 cm²/V · s. (Refer to Chapter 7, *Anderson and Anderson*)

Q10. A good way to check the validity of a derivation is to verify that it reduces to the expected result for a particular known case. For example, in the simple model we neglected velocity saturation, and in the later model we took it into account. Since the high field that causes velocity saturation occurs in shorter channel devices, we would expect that the later model would reduce to the simple model for long-channel devices.

- a) Show that in the limit of large L , V_{DSsat} has given by Equation (7.68, *Anderson and Anderson*)

reduces to $V_{DSsat} = (V_{GS} - V_T)$ as given by Equation (7.26, *Anderson and Anderson*) for the simple (long-channel) model .

You may find the following information useful:

For $x < 1$

$$(1 \pm x)^n = 1 \pm nx + \frac{n(n-1)x^2}{2!} \pm \dots$$

$$(1 \pm x)^{-n} = 1 \mp nx + \frac{n(n+1)x^2}{2!} \mp \dots$$

b) Show that in the limit of large L , I_{Dsat} as given by Equation (7.6, *Anderson and Anderson*) reduces to that of Equation (7.25, *Anderson and Anderson*) for the simple model.

Q11. Find an approximate closed-form expression for the drain current, accounting for series resistance and velocity saturation. That is, solve for I_D in Equation (7.72, *Anderson and Anderson*). Neglect the term in I_D .

Q12. In the MOS process, structures like the gate of a transistor are used to make capacitors as well. If the oxide thickness is 4 nm, what area is needed to achieve a capacitance of 1 pF? The permittivity of silicon dioxide is $3.9\epsilon_0$.

Q13. For Semiconductor-Oxide-Semiconductor (SOS) capacitor. Answer the following based on these cases:

- Semiconductor is n-type
- Semiconductor is intrinsic
- Semiconductor is n+ polysilicon
- Draw the energy band diagram for the structure @ $V_g > 0$ and very large
- Draw the charge profile @ $V_g > 0$ and very large
- Plot the CV characteristics of SOS capacitor

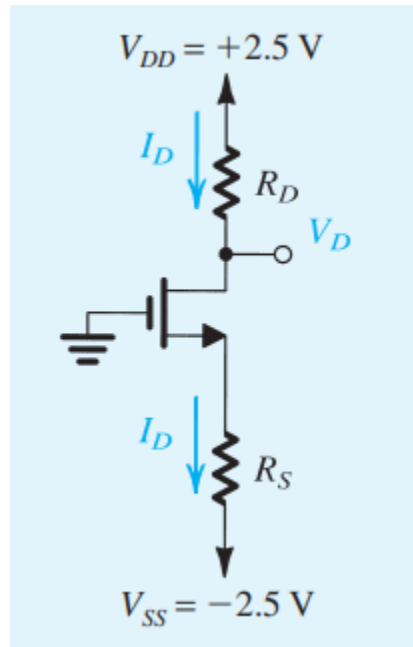
Q14. Why does the drain current saturate after pinch-off in a MOSFET? Additionally, what is the condition of the substrate (p-type, depletion, or inversion) near the source and drain regions under pinch-off?

Q15. Derive the expressions for the subthreshold slope, threshold voltage, and drain current in the off, linear, and saturation regions of a MOSFET. Clearly specify the corresponding V_{gs} and V_{ds} operating ranges for each region?

Q16. List all MOSFET design parameters control the Area, Power and Speed of the MOSFET? Which parameters have limitations in design and which parameter is most useful for designing better performing MOSFET?

Q17. What is the threshold voltage dilemma of the MOSFET? What happens when V_{th} increases or decreases to speed and power?

Q18. Design the circuit of the figure below: that is, determine the values of R_D and R_S so that the transistor operates at $I_D = 0.4 \text{ mA}$ and $V_D = +0.5 \text{ V}$. The NMOS transistor has $V_t = 0.7 \text{ V}$, $\mu_n C_{ox} = 100 \text{ } \mu\text{A/V}^2$, $L = 1 \text{ } \mu\text{m}$, and $W = 32 \text{ } \mu\text{m}$. Neglect the channel-length modulation effect (i.e., assume that $\lambda = 0$).



Q19. For the circuit in Figure , find the value of R that results in $V_D = 0.7 \text{ V}$. The MOSFET has $V_{th} = 0.5 \text{ V}$, $\mu_n C_{ox} = 0.4 \text{ mA/V}^2$, $W/L = 0.72 \text{ } \mu\text{m} / 0.18 \text{ } \mu\text{m}$, and $\lambda = 0$?

